8. ESTIMATION OF UREA BY DIACETYL MONOXIME METHOD

Aim:

To estimate the amount of urea present in the given test solution

Principle:

Under acidic conditions when urea is heated with compounds containing two adjacent carbonyl groups such as diacetylmonoxime coloured products are formed. On heating, it decomposes to give hydroxylamine and diacetyl, which then condenses with urea to give rose purple diazinederivative.The colour intensity is measured colorimetrically at 520nm.

Reagents:

1. **Acid reagent:Reagent A:** 5g of ferric chloride is dissolved in 20ml water and 100ml of orthophosphoric acid (85%) is slowly added with stirring and the volume is made up to 250ml with water;Reagent B:200ml of conc. sulphuric acid is added to 800ml water;0.5ml of reagent A is added to 1 litre of Reagent B.
2. **Colour Reagent:** Reagent C: Dissolve 20g diacetylmonoxime in 1 litre of water;Reagent D: Dissolve 5g thiosemicarbazidein l litre of water;Mixed 67ml of C and 67ml of D and made this volume upto1 litre with distilled water.
3. **Urea Stock solution:** Dissolve 100mg Urea in 100 ml of distilled water.
4. **Working standard solution:** 10ml of the stock is diluted to 100ml using distilled water.

   Concentration = 100µg / ml.

Procedure:

0.2-1.0ml of working standard urea solution is pipetted out into five different test tubes and labelled as S1-S5 with the concentration of 20-100µg. 1.0ml of the test solution is taken in a test tube and labelled as T_1. The volume in all the test tubes is made up to 2.0ml with distilled water. 2.0ml of water is taken as a blank solution. 2.0ml of colour reagent and 2.0ml of acid reagent are added to all the tubes and placed in a boiling water bath for 20 min. The tubes are cooled and pink colour developed is read at 520nm.

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